EXERCISE 6.8

1. Show that the straight lines $\vec{r} = (5\hat{i} + 7\hat{j} - 3\hat{k}) + s(4\hat{i} + 4\hat{j} - 5\hat{k})$ and $\vec{r} = \left(8\hat{i} + 4\hat{j} + 5\hat{k}\right) + t\left(7\hat{i} + \hat{j} + 3\hat{k}\right)$ are coplanar. Find the vector equation of the plane in which they lie.
2. Show that the lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{3}$ and $\frac{x-1}{-3} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar. Also, find the plane containing these lines.
3. If the straight lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{m^2}$ and $\frac{x-3}{1} = \frac{y-2}{m^2} = \frac{z-1}{2}$ are coplanar, find the distinct real values of m.
4. If the straight lines $\frac{x-1}{2} = \frac{y+1}{\lambda} = \frac{z}{2}$ and $\frac{x+1}{5} = \frac{y+1}{2} = \frac{z}{\lambda}$ are coplanar, find λ and equations of the planes containing these two lines.

6.8.10 Angle between two planes

The angle between two given planes is same as the angle between their normals.

Theorem 6.18

The acute angle θ between the two planes $\vec{r} \cdot \vec{n}_1 = p_1$ and $\vec{r} \cdot \vec{n}_2 = p_2$ is given by

$$\theta = \cos^{-1}\left(\frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}\right)$$

Proof

If θ is the acute angle between two planes $\vec{r} \cdot \vec{n}_1 = p_1$ and $\vec{r} \cdot \vec{n}_2 = p_2$, then θ is the acute angle between their normal vectors $\vec{n}_1$ and $\vec{n}_2$.

Therefore, $\cos\theta = \left(\frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}\right) \Rightarrow \theta = \cos^{-1}\left(\frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}\right)$ ... (1)

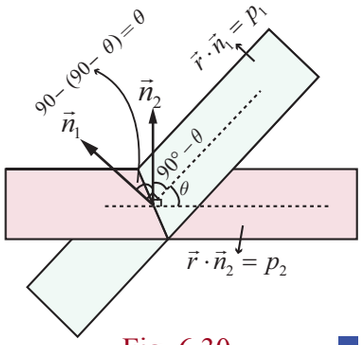


Fig. 6.30

Remark

(i) If two planes $\vec{r} \cdot \vec{n}_1 = p_1$ and $\vec{r} \cdot \vec{n}_2 = p_2$ are perpendicular, then $\vec{n}_1 \cdot \vec{n}_2 = 0$

(ii) If the planes $\vec{r} \cdot \vec{n}_1 = p_1$ and $\vec{r} \cdot \vec{n}_2 = p_2$ are parallel, then $\vec{n}_1 = \lambda\vec{n}_2$, where λ is a scalar

(iii) Equation of a plane parallel to the plane $\vec{r} \cdot \vec{n} = p$ is $\vec{r} \cdot \vec{n} = k$, $k \in \mathbb{R}$

Theorem 6.19

The acute angle θ between the planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ is given by $\theta = \cos^{-1}\left(\frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}\right)$

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